

Piyush Paul

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Professional Summary

Computer Science undergraduate with hands-on experience building AI-powered SaaS platforms, cybersecurity tools, and high-performance backend systems. Proficient across the full stack — from low-level C/DPDK network programming to cloud-deployed agentic AI pipelines. Known for calm, analytical problem-solving under pressure, strong pattern-recognition instincts, and a relentless drive to ship working systems end-to-end. Particularly interested in autonomous AI agents and offensive/defensive cybersecurity.

Technical Skills

- **LLMs & Agentic AI:** Multi-agent orchestration, prompt engineering, RAG, LLM fine-tuning concepts, Gemini, OpenAI GPT, Anthropic Claude
- **Backend & APIs:** Python (FastAPI, Flask), REST API design, microservices, SSE architecture
- **Cloud & DevOps:** GCP, Docker, Git, CI/CD
- **ML & AI:** Agentic workflows, LLM integration, NLP, autonomous pipeline design
- **Databases:** MongoDB, SQL, SQLite, Supabase
- **Languages:** Python, JavaScript, TypeScript, C, C++, Java, SQL, Rust

Experience

Backend Developer Intern — Fluencer Market (Remote)

Dec 2024 – Apr 2025

- Designed and developed Flask REST APIs supporting influencer-brand collaboration workflows
- Reduced backend API latency by 35% through optimized database queries and response caching
- Delivered scalable backend services collaborating with cross-functional engineering teams

Projects

ARTA — Autonomous Red Team Agent (Multi-Agent AI)

[GitHub](#) | Apr 2026

FastAPI, Gemini AI, SQLite, Puppeteer, Nmap, Metasploit

- Built an AI-driven autonomous penetration-testing pipeline using Gemini — a multi-agent system that coordinates target discovery, CVE analysis, and exploit validation sub-agents in hierarchical workflows
- Implemented Server-Sent Events (SSE) architecture for real-time asynchronous log streaming and live findings visualization
- Integrated LLM orchestration with external tooling (Nmap for scanning, Metasploit for exploitation), maintaining tight context control across agent handoffs
- Won MLH Hack Days — validates ability to ship production-quality AI agent systems

AstroRoni — Full-Stack AI SaaS Platform (Cloud / ML Integration)

[Project Link](#) | Sep 2025 – Apr 2026

Next.js, Flask, MongoDB, Razorpay, OpenAI, GCP

- Built and deployed a scalable cloud-based microservices architecture on GCP handling authentication, payments, and AI-driven report generation using OpenAI APIs
- Designed RESTful APIs and data models supporting concurrent multi-user LLM-powered workflows
- Integrated LLM inference into a production pipeline with payment gating and async result delivery

PacketForge — High-Performance Systems Pipeline (Performance Engineering)

May 2026

C, DPDK, Multi-threading, FastAPI, Next.js, QEMU/KVM

- Architected a dual-NIC multi-threaded packet-processing pipeline sustaining 470K pps
- Built a real-time telemetry dashboard (FastAPI + React) for live performance monitoring
- 1st Runner-Up at Synchronicity S2 (24-hour hackathon)

Achievements

- **MLH Hack Days — Winner (2026):** Built ARTA, an AI-powered autonomous multi-agent penetration-testing system ([GitHub](#))
- **Synchronicity S2, Jadavpur University — 1st Runner-Up (2026):** Built PacketForge, a DPDK-based DDoS scrubber
- **MLH HackTropica — 1st Runner-Up (2026):** Built NeighbourNet, an offline mesh-networking solution for disaster communication

Education

JIS College of Engineering, West Bengal, India

Jun 2025 – Present

Bachelor of Technology in Computer Science (AI/ML) — SGPA: 7.9/10

Relevant coursework: Data Structures & Algorithms, Operating Systems, DBMS